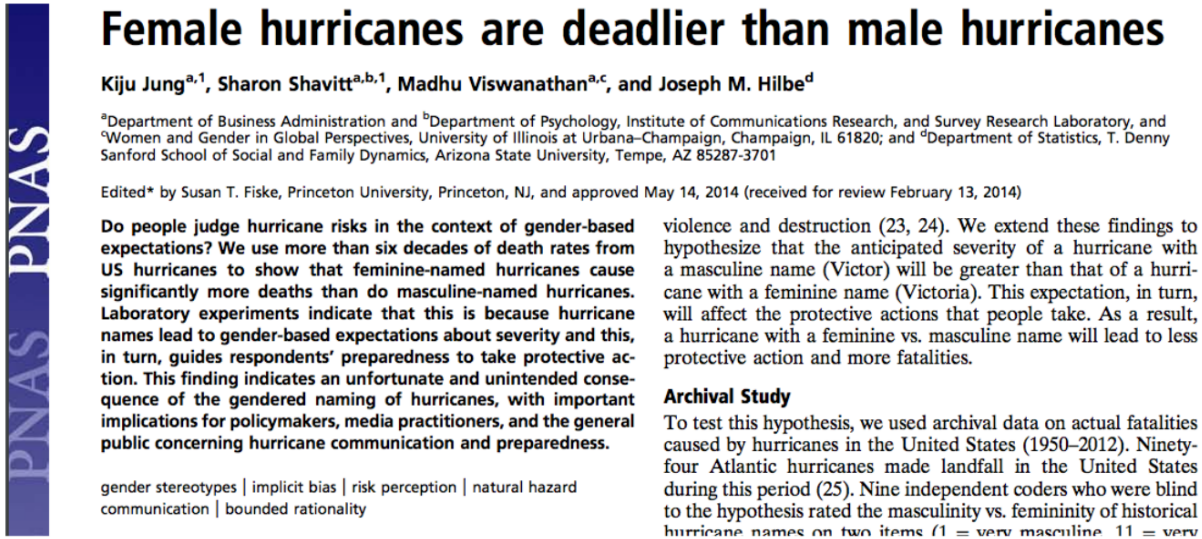
Lecture 15: More on multiple regression and causality

- We have seen examples, where additional knowledge was gained from multiple regression used wisely.
- Remember that interpreting the (regression) parameter estimates always depends upon what you believe the causal model, encoded as a DAG for us, is.
- In this last lecture of the week, we are going to see more examples of controls (conditioning) and their effects. Desired. Or not?

Some issues with causality



PNAS

Female hurricanes are deadlier than male hurricanes

Kiju Jung[a,1], Sharon Shavitt[a,b,1], Madhu Viswanathan[a,c], and Joseph M. Hilbe[d]

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Edited* by Susan T. Fiske, Princeton University, Princeton, NJ, and approved May 14, 2014 (received for review February 13, 2014)

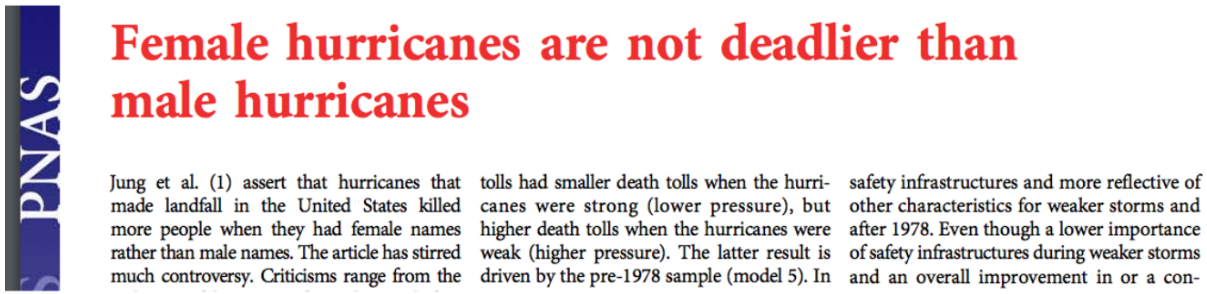
Do people judge hurricane risks in the context of gender-based expectations? We use more than six decades of death rates from US hurricanes to show that feminine-named hurricanes cause significantly more deaths than do masculine-named hurricanes. Laboratory experiments indicate that this is because hurricane names lead to gender-based expectations about severity and this, in turn, guides respondents' preparedness to take protective action. This finding indicates an unfortunate and unintended consequence of the gendered naming of hurricanes, with important implications for policymakers, media practitioners, and the general public concerning hurricane communication and preparedness.

gender stereotypes | implicit bias | risk perception | natural hazard communication | bounded rationality

violence and destruction (23, 24). We extend these findings to hypothesize that the anticipated severity of a hurricane with a masculine name (Victor) will be greater than that of a hurricane with a feminine name (Victoria). This expectation, in turn, will affect the protective actions that people take. As a result, a hurricane with a feminine vs. masculine name will lead to less protective action and more fatalities.

Archival Study

To test this hypothesis, we used archival data on actual fatalities caused by hurricanes in the United States (1950–2012). Ninety-four Atlantic hurricanes made landfall in the United States during this period (25). Nine independent coders who were blind to the hypothesis rated the masculinity vs. femininity of historical hurricane names on two items (1 = very masculine, 11 = very



PNAS

Female hurricanes are not deadlier than male hurricanes

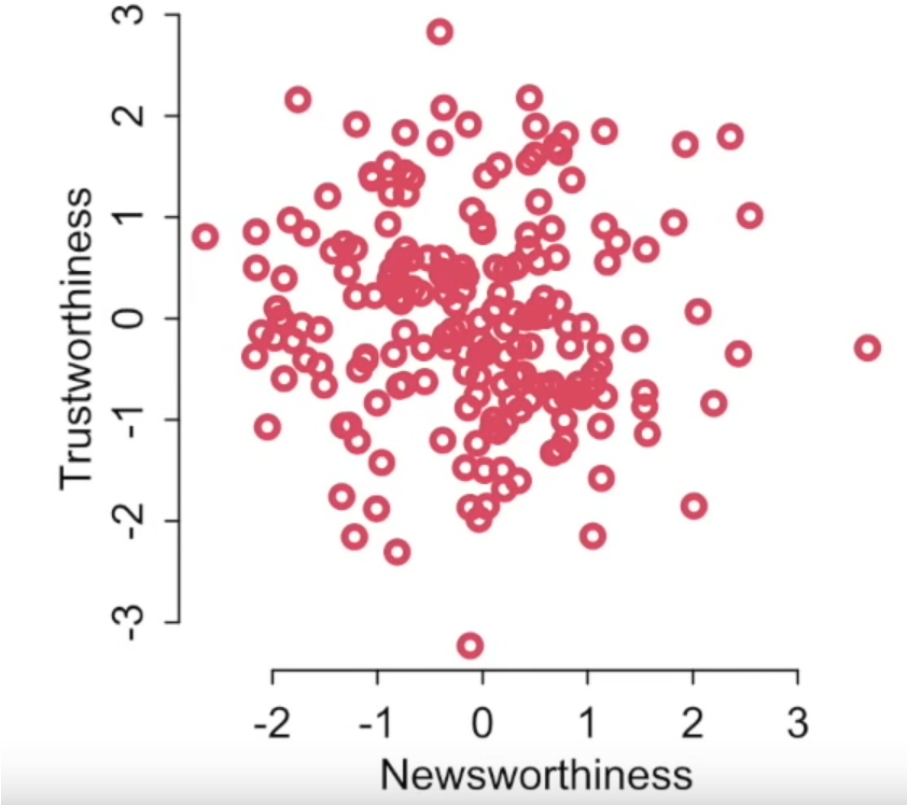
Jung et al. (1) assert that hurricanes that made landfall in the United States killed more people when they had female names rather than male names. The article has stirred much controversy. Criticisms range from the tolls had smaller death tolls when the hurricanes were strong (lower pressure), but higher death tolls when the hurricanes were weak (higher pressure). The latter result is driven by the pre-1978 sample (model 5). In safety infrastructures and more reflective of other characteristics for weaker storms and after 1978. Even though a lower importance of safety infrastructures during weaker storms and an overall improvement in or a con-

```
data(Hurricanes)
```

How is this related to multiple regression?

Selection-distortion effect

aka Berkson's paradox



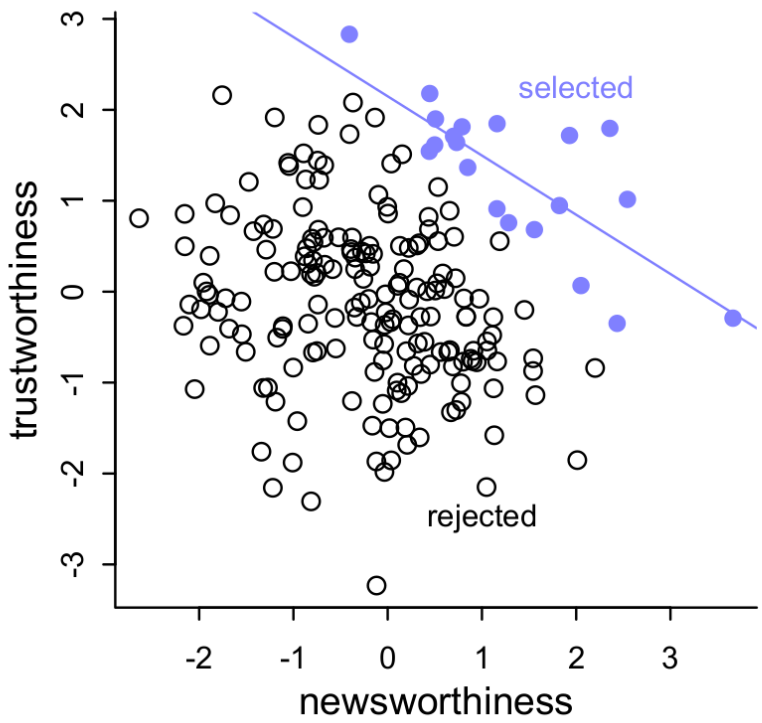
(simulation at the start of Chapter 6 of the SR book)

- Some biases arise from selection.
- Suppose: 200 grant applications.
- Each is scored on newsworthiness and trustworthiness.
- These 2 scores are not associated in the population.

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Selection-distortion effect

aka Berkson's paradox

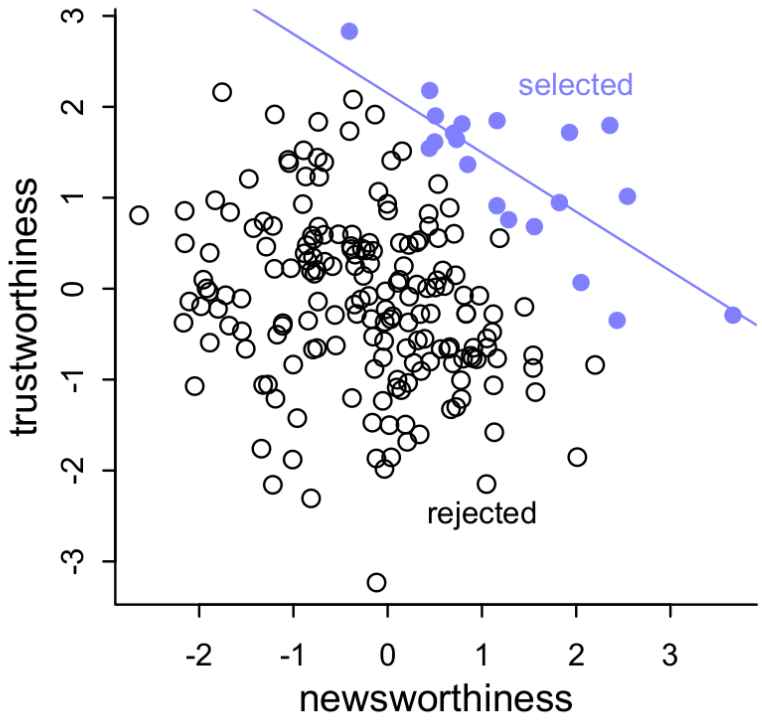


- Some biases arise from selection.
- Suppose: 200 grant applications.
- Each is scored on newsworthiness and trustworthiness.
- These 2 scores are not associated in the population.
- Strong (negative?!) association after selection.

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Selection-distortion effect

is an example of a collider effect



- Awarded grants need be sufficiently newsworthy (N) and trustworthy (T).
- Few grants are high in both.
- The negative association is a result of award (A).

$$N \longrightarrow A \longleftarrow T$$

- And we only see those who succeed usually.

Regression as a (wicked) oracle

- Regression automatically focuses in the most informative cases.
- Cases that don't help are automatically ignored.
- But the oracle is neither kind nor subtle, as carefully!

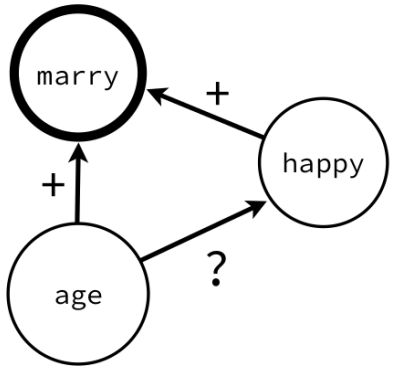
- You could just add all the available predictors: "we controlled for..." is so often encountered.
- Almost always a bad idea: creates confounds, residual confounding and over-fitting (see Week 7).

Endogeneous colliders

- Similar other examples: (i) restaurants in good locations have bad food or (ii) Attractive actors are less skilled.
- Collider bias can arise through statistical processing.
- Endogeneous selection: if you condition on a (aka stratify by) a collider, it creates a phantom non-causal associations.
- Said differently, conditioning on a collider is going to select a sub-population.
- Example: does age influence happiness? Work through the example and the code (Section 6.3.1).

Age and Happiness

- Estimand: estimate of age on happiness.
- Possible confound: marital status.
- Suppose (provocatively) that age has no influence on happiness.
- But both age and happiness influence marital status,

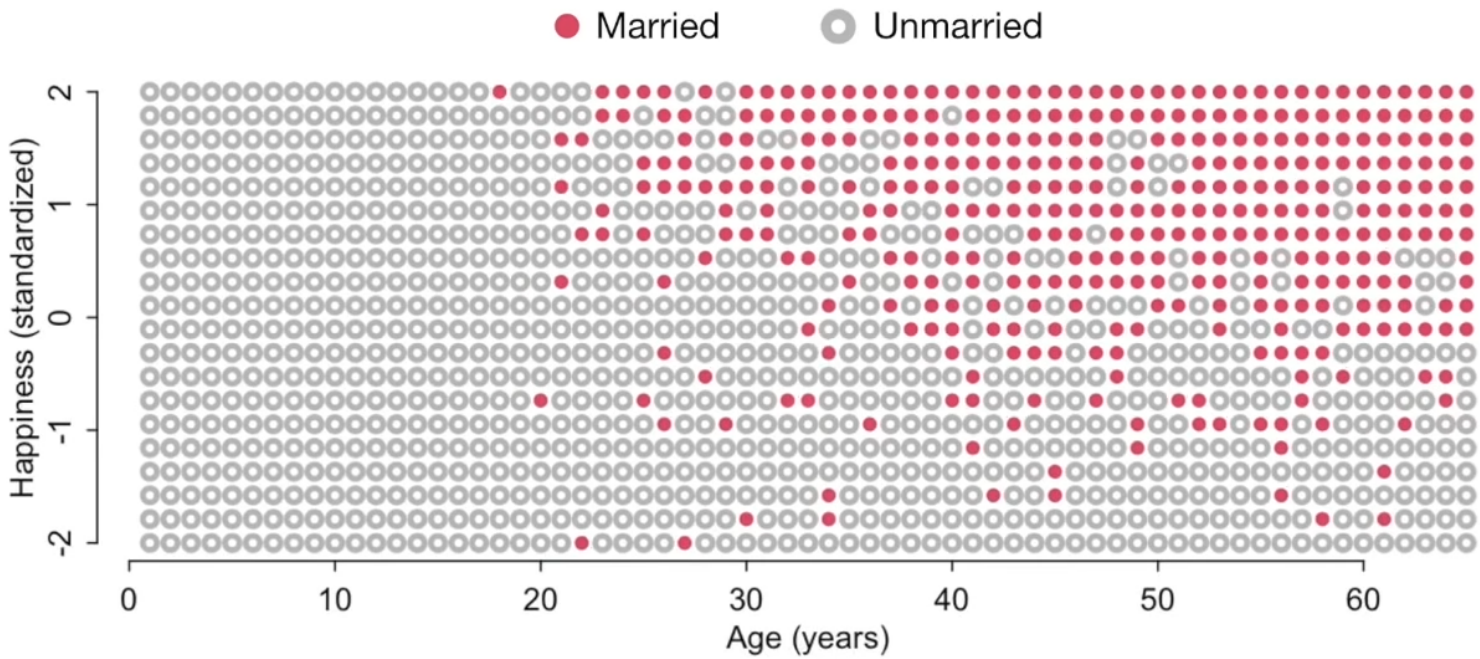



```
d <- sim_happiness(seed=1977,
  N_years = 1000)
precis(d)
```

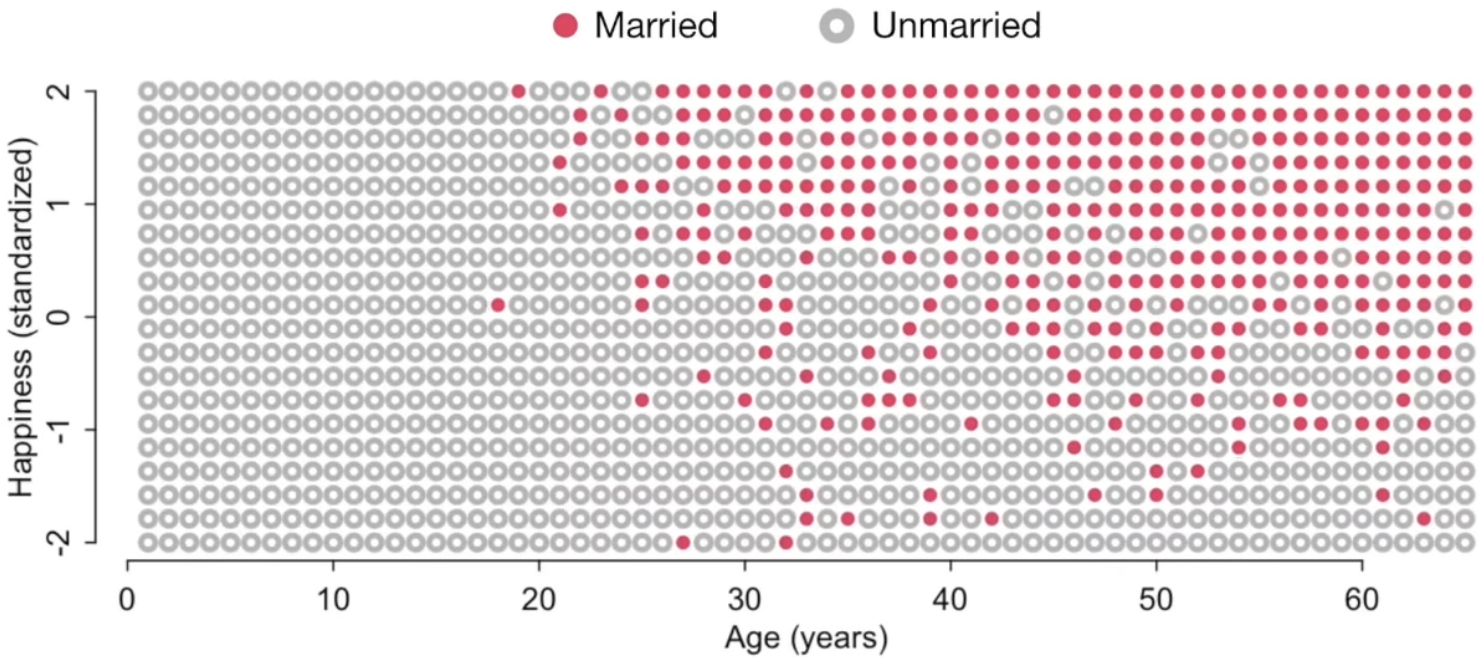
```
'data.frame': 1300 obs. of 3 variables:
          mean    sd  5.5% 94.5%     histogram
age       33.0 18.77  4.00 62.00
married    0.3  0.46  0.00  1.00
happiness  0.0  1.21 -1.79  1.79
```

- 20 people born each year.
- Uniform happiness at birth, never changes.
- At 18, $P(\text{marry}) \propto \text{happy}$ each year.
- Married until death. At 65, everyone disappears (to a sunny place :)).

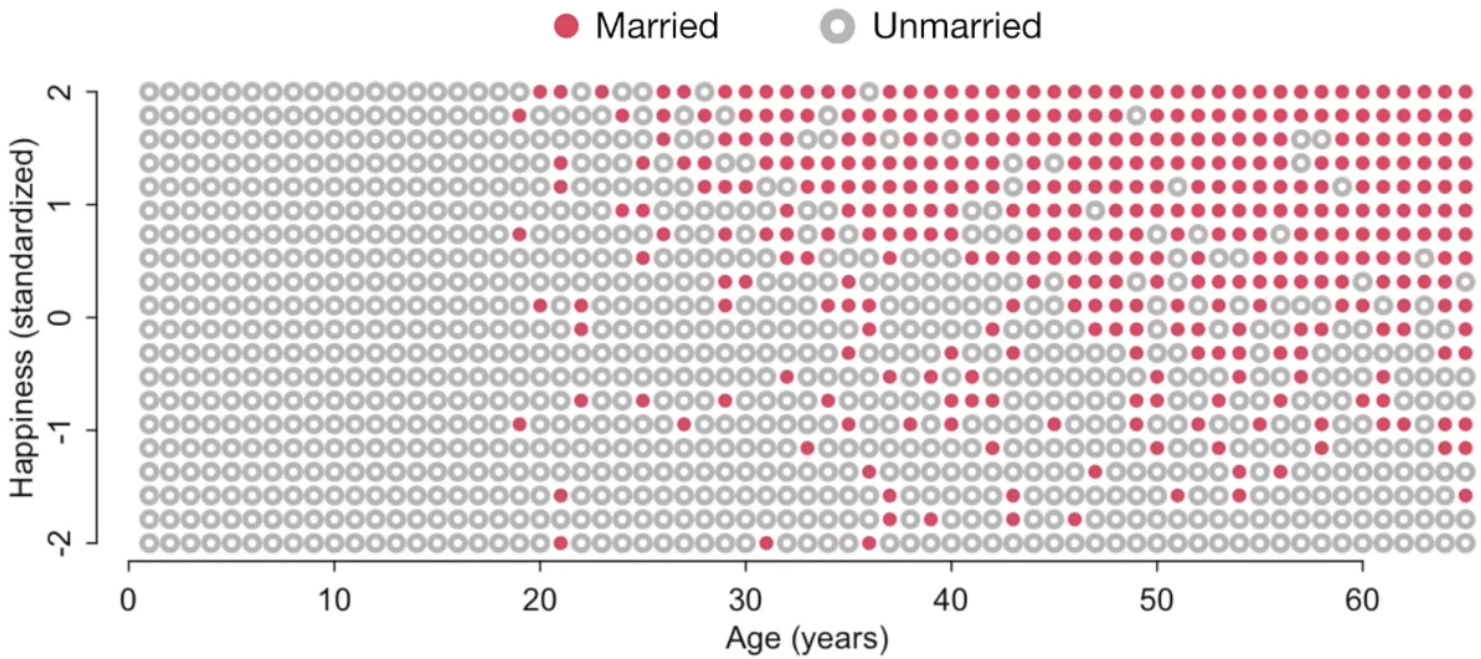
Dynamical simulation of age, happiness and marriage



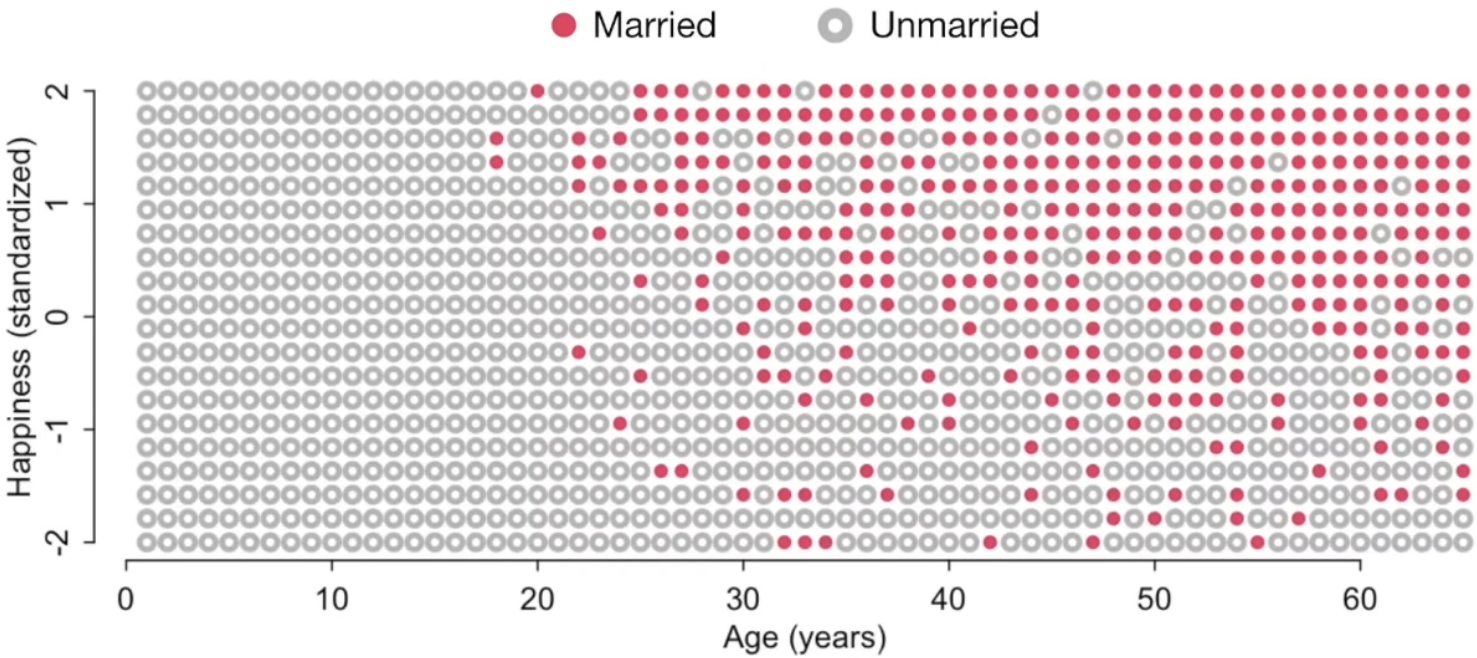
Dynamical simulation of age, happiness and marriage



Dynamical simulation of age, happiness and marriage



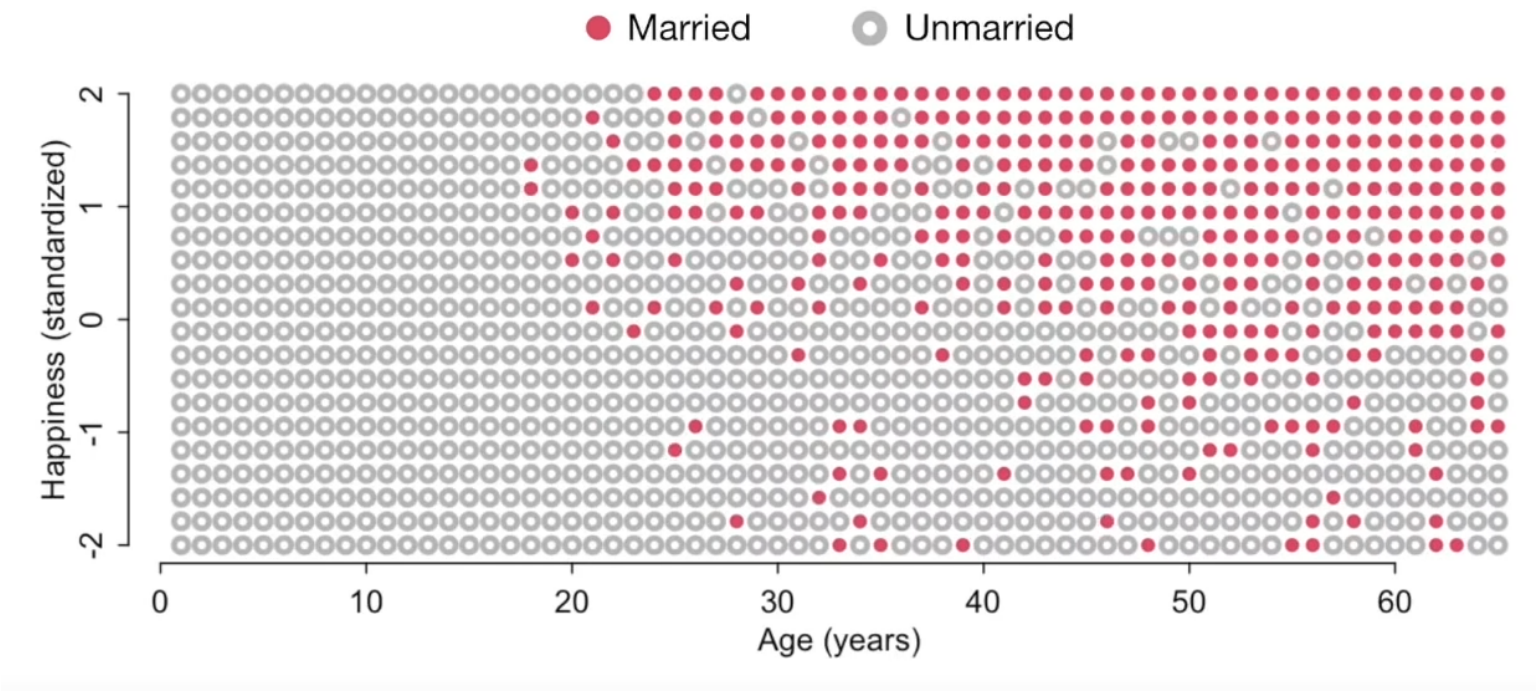
Dynamical simulation of age, happiness and marriage



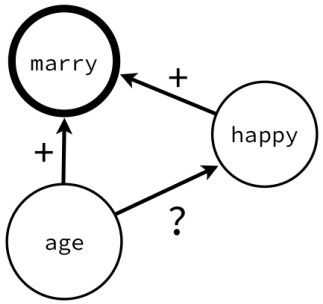
Dynamical simulation of age, happiness and marriage

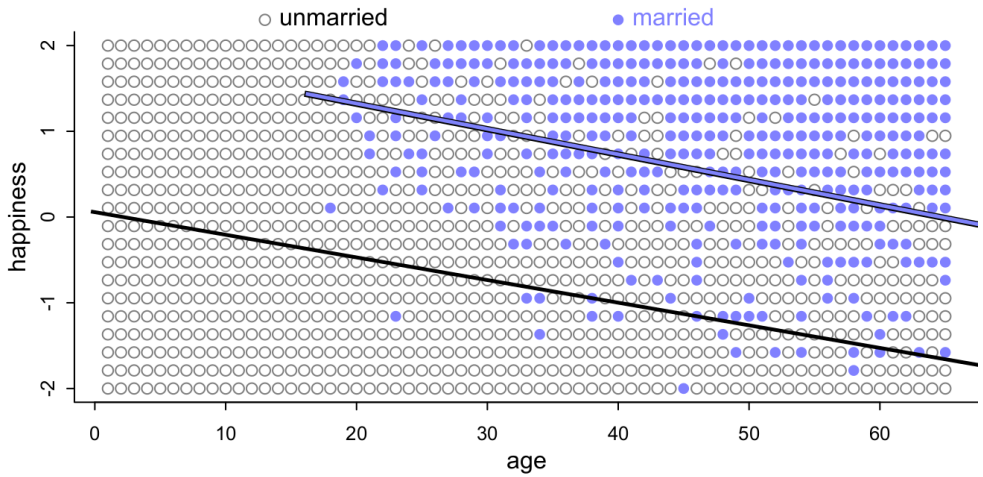


Dynamical simulation of age, happiness and marriage



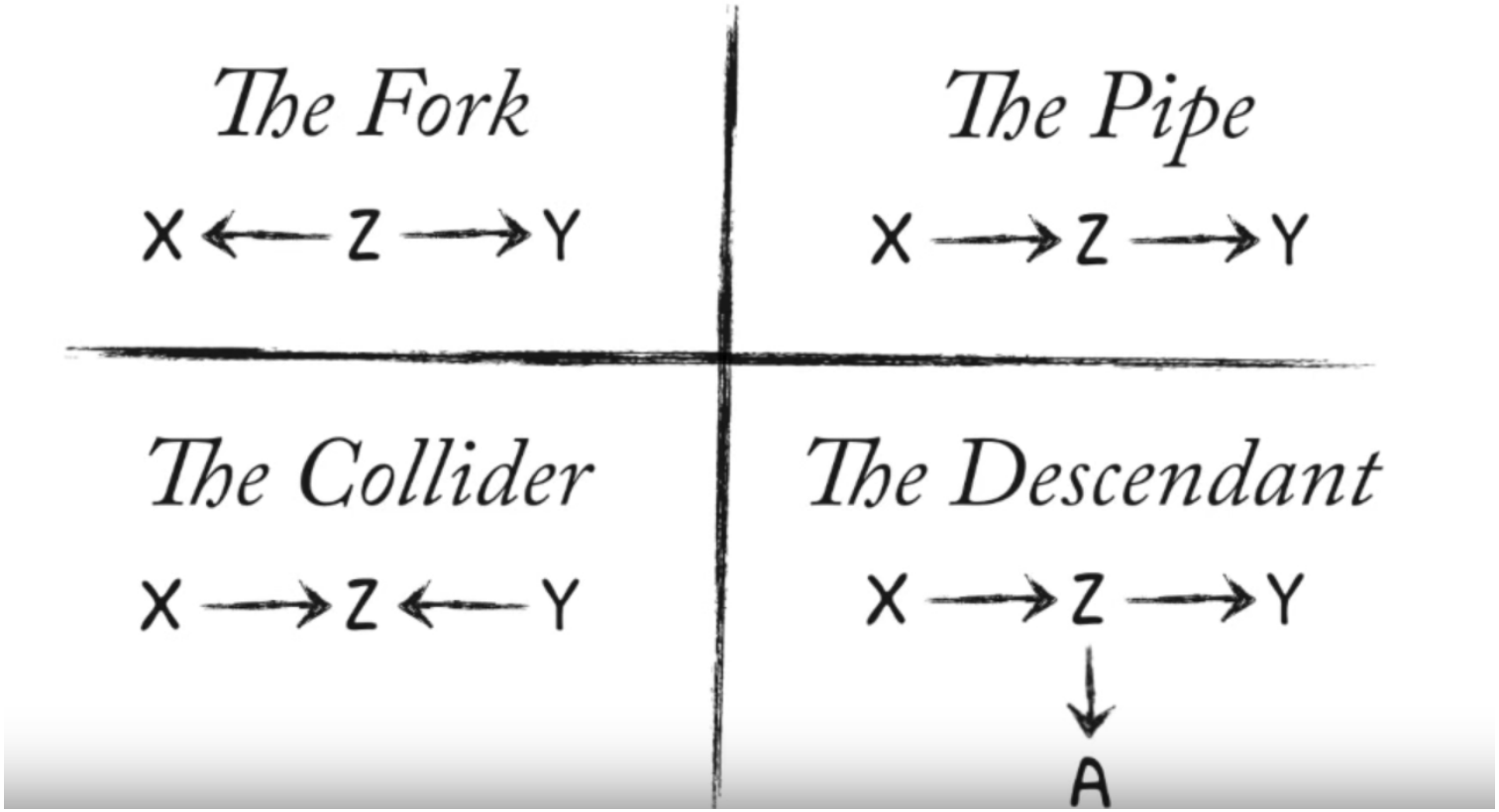
- Are older people less happy? Controlling for the marriage status creates the confound.
- We cannot know whether we should control for a variable without a causal model.





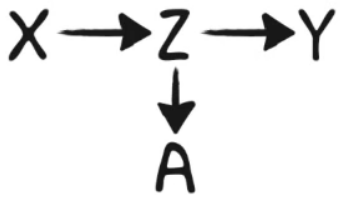
Back to the causal cutlery

One more to explore



The "descendant"

A is the descendant



- How descendants behave depends upon what they are attached to.
- Above, there is a pipe (could be any other 3 other blocks).
- Conditioning on *A* has the same effect as conditioning on *Z*.
- Here, *X* and *Y* are causally associated through *Z*. *A* holds information about *Z*.
- Once stratified by *A*, *X* and *Y* are less associated.

(Same principle if we have a fork. If *Z* was a collider, conditioning on *A* would make *X* and *Y* more associated than they really are)

The descendant: a binary example

Without/with stratification - pipe case

```
n <- 1000
X <- rbern(n, 0.5)
Z <- rbern(n,(1-X)*0.2+X*0.8)
Y <- rbern(n,(1-Z)*0.1+Z*0.9)
A <- rbern(n, (1-Z)*0.05+Z*0.95)

table(X,Y)
   Y
X     0   1
  0 356 148
  1 128 368
cor(X,Y)
[1] 0.4485
```

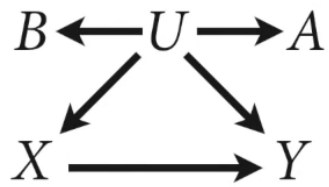
```
table(X[A==0], Y[A==0])
     0   1
0  330  67
1   88  28
cor(X[A==0], Y[A==0])
[1] 0.07819736

table(X[A==1], Y[A==1])

     0   1
0   26  81
1   40 340
cor(X[A==1], Y[A==1])
[1] 0.1666063
```

Descendants are everywhere

- Lots of the measurements we take are **proxies** of what we want to measure.
- It would make sense not to draw graphs with real ideal measurements we wish we had, but scientific models that measurement the proxies we have access to.
- Factor Analysis (see 161.250).
- Measurement error models.
- Social networks: we measure behaviour on the network, not the network itself.

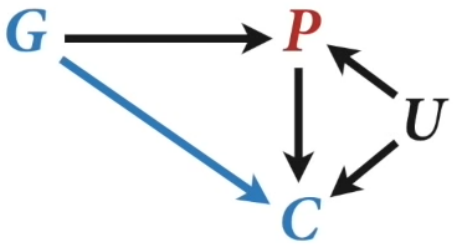


U: Unobserved confound

We would want the effect of X and Y as estimand. confounded by U we do not observe. A and B can help. Maybe not directly in the model.

Unobserved confounds

- Unmeasured confounds (U) can exist and they can ruin your day.
- Estimand: direct effect of G on C.
- It means you need to block the $G \rightarrow P \rightarrow C$ pipe.
- But what happens when we condition on P?



Conclusions

- If you don't have a causal model, you cannot make inferences from a multiple regression.
- The regression itself does **not** provide the evidence you need to justify a causal model.
- Instead you need some science.

- Have a good weekend and see you to talk about a haunted DAG :)